

Zomato Restaurant Data Analysis

Detailed Documentation Report — Python / Pandas / NumPy / Matplotlib / Seaborn

1. Project Overview

This project performs Exploratory Data Analysis (EDA) on a Zomato restaurant dataset. The assignment requires dataset exploration, basic cleaning and preprocessing, descriptive statistics, business-oriented comparisons, five specified visualizations, and a final interpretation of the findings.

The notebook implements the workflow in sequence: library setup, dataset loading, ID normalization, structure inspection, missing-value analysis, duplicate checks, type conversion, statistical analysis, business comparisons, visualization, an additional correlation heatmap, and final conclusions.

2. Dataset Overview

- Dataset file used by notebook: test_zomato.csv
- Rows: 9,552; Columns: 22
- Unique cities: 142
- Unique restaurant names: 7,447
- Most represented city: New Delhi — 5,473 restaurants
- Most frequent cuisine after splitting cuisine lists: North Indian — 3,960 listings

Key fields used include Restaurant Name, City, Cuisines, Average Cost for two, Has Online delivery, Has Table booking, Price range, Aggregate rating, Rating text, and Votes.

3. Data Loading and Preparation

The notebook searches the Kaggle input directory for a CSV/XLSX file and loads a CSV using `pandas.read_csv` with latin-1 encoding. An existing 'Unnamed' column is renamed to ID; otherwise a sequential ID is inserted. Column names are stripped of surrounding whitespace.

4. Data Cleaning

No full duplicate rows and no duplicate Restaurant IDs were found. The dataset remained at 9,552 rows after duplicate removal.

Average Cost for two is cleaned by removing commas and converting to numeric. Aggregate rating, Votes, and Price range are converted to numeric. Missing Cuisines values are filled with 'Not Specified', and City values are trimmed.

5. Statistical Analysis and Core Results

- Average rating — all rows: 2.67.
- Average rating — rated only: 3.44.
- Average cost for two: 1,199.33; median: 400.00.
- Highest-voted restaurant: Toit, Bangalore — 10,934 votes, rating 4.8.
- Unique cities: 142.

- Highest restaurant presence: New Delhi — 5,473.
- Most frequent cuisine: North Indian — 3,960.

6. Business Comparisons

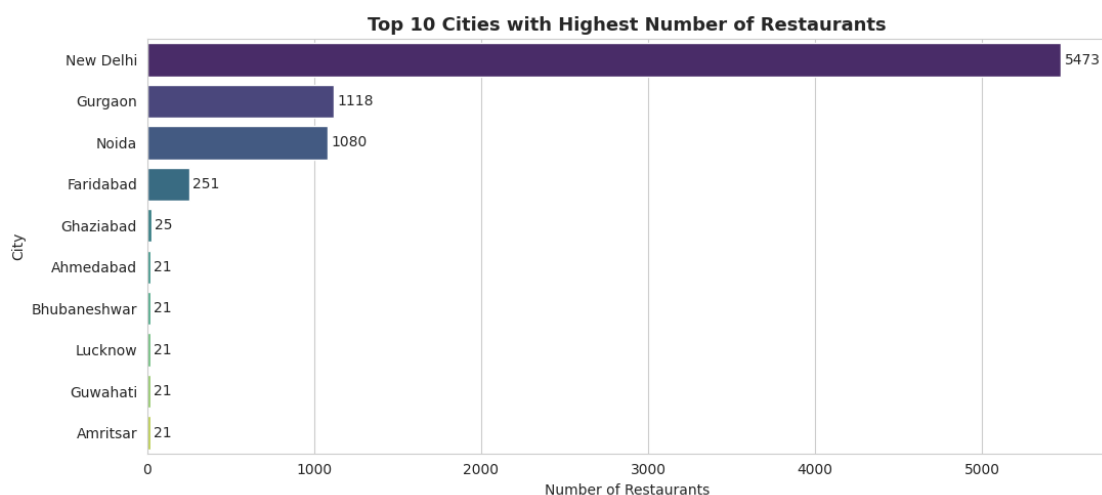
Online delivery: among rated restaurants, Yes = 3.38 and No = 3.47. Thus online delivery is not associated with a higher average rating in this dataset.

Table booking: among rated restaurants, Yes = 3.59 and No = 3.41. This is an association and should not be interpreted as causal.

Price range: average ratings are 3.24, 3.38, 3.78 and 3.89 for price ranges 1–4 respectively.

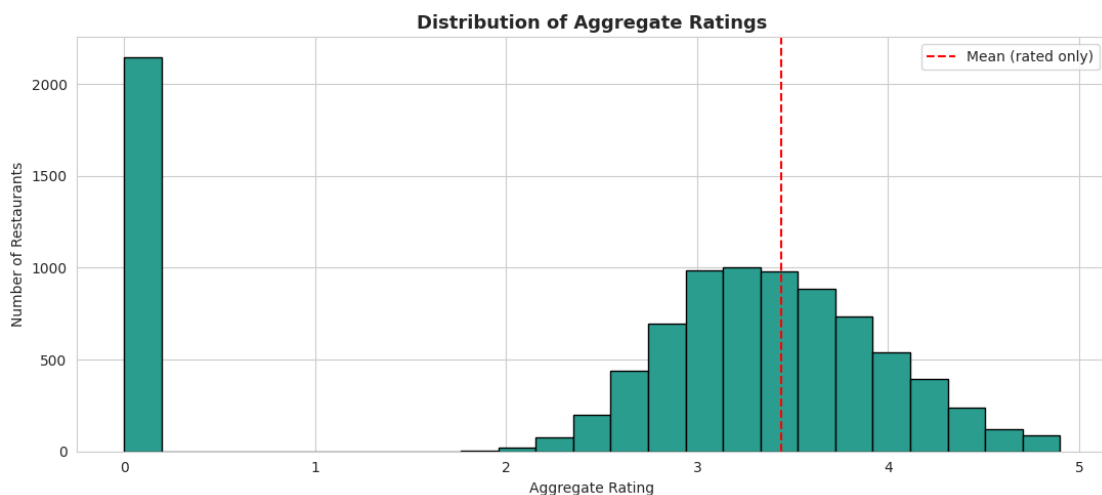
7. Required Visualizations

Figure 1 — Top 10 Cities



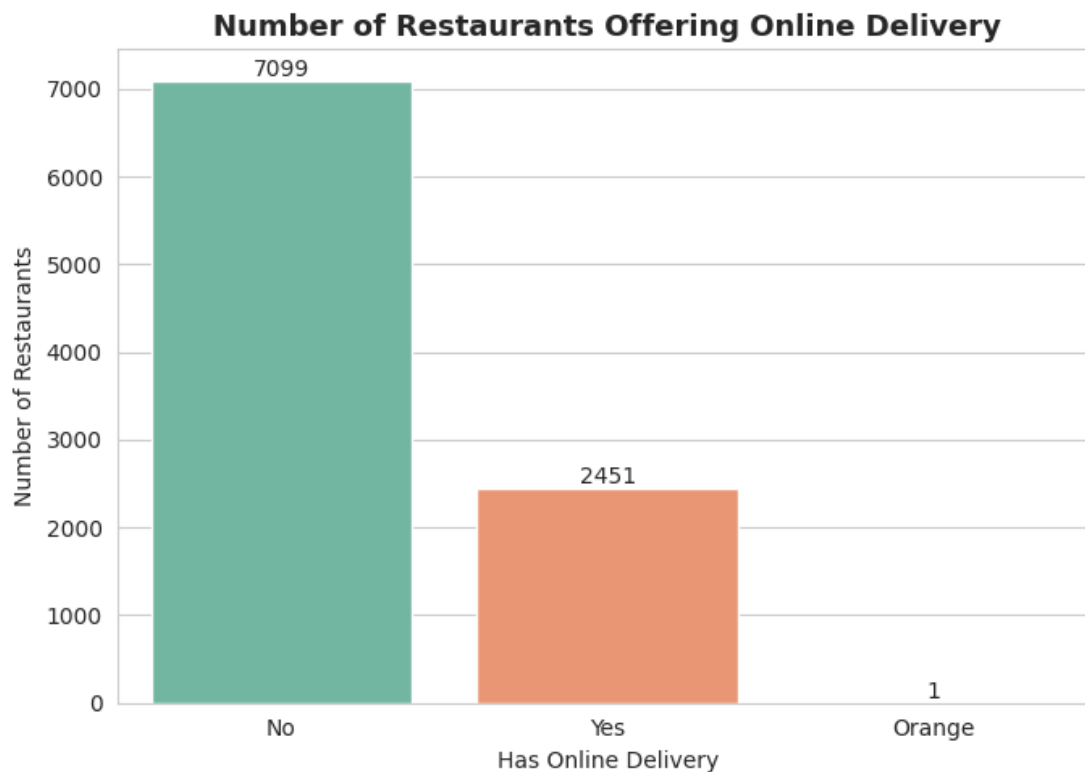
New Delhi dominates the dataset; Gurgaon and Noida follow.

Figure 2 — Distribution of Ratings



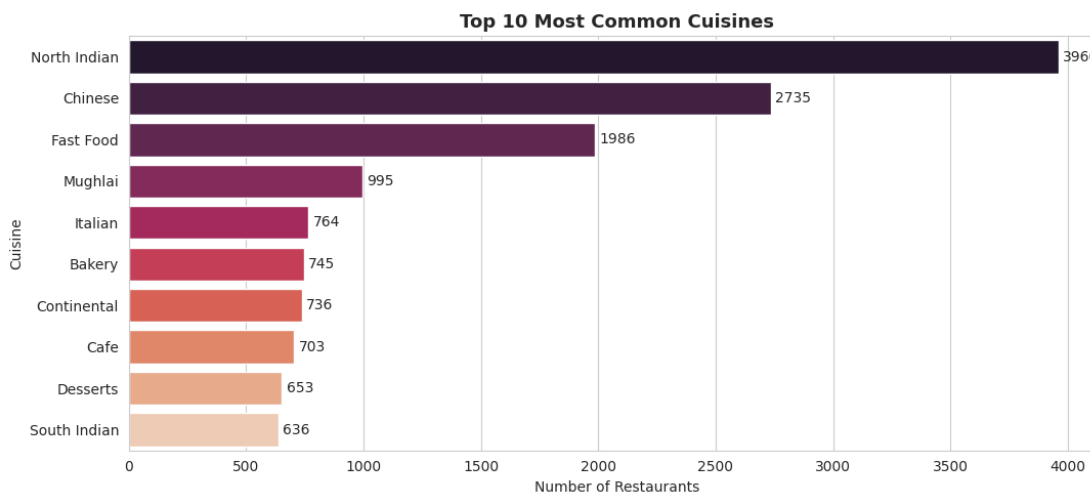
A large rating-0 spike represents unrated records; rated scores cluster mainly around 3–4.

Figure 3 — Online Delivery Availability



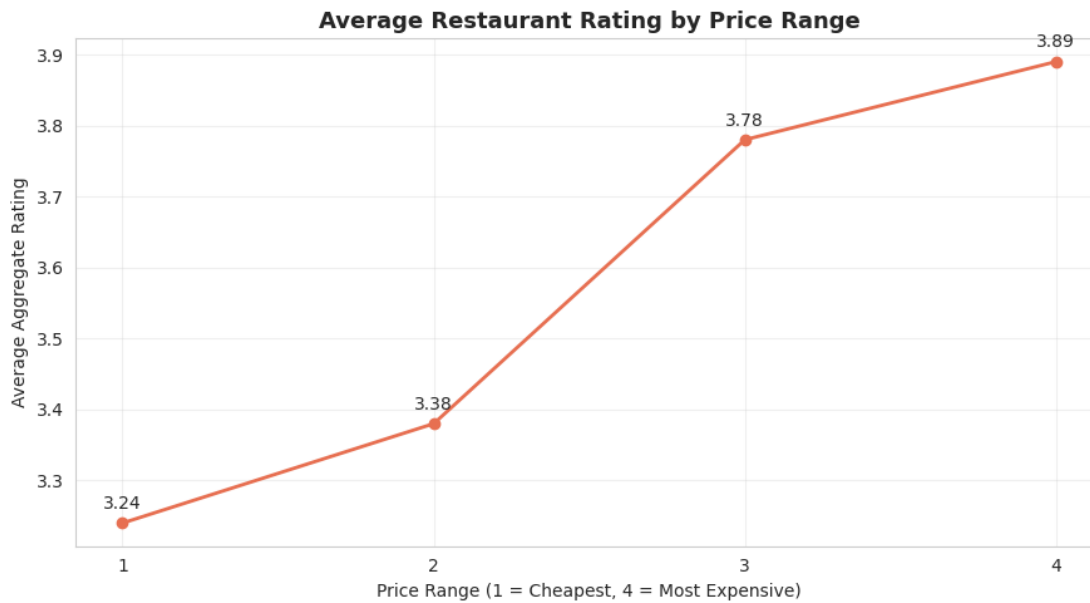
7,099 are No and 2,451 are Yes; one anomalous Orange value is also surfaced.

Figure 4 — Top 10 Cuisines



North Indian is the most frequent cuisine listing.

Figure 5 — Average Rating by Price Range



Average rating rises consistently as price range increases.

8. Insights and Conclusion

- New Delhi has the highest restaurant presence with 5,473 listings.
- North Indian is the most frequently listed cuisine with 3,960 listings.
- Online delivery restaurants have a slightly lower rated-only average (3.38) than non-delivery restaurants (3.47).
- Table-booking restaurants have a higher rated-only average (3.59) than restaurants without booking (3.41).
- Price range shows the clearest monotonic relationship with rating: $3.24 \rightarrow 3.38 \rightarrow 3.78 \rightarrow 3.89$.
- Zero ratings should be interpreted as unrated rather than poor ratings when calculating meaningful rating averages.

9. Limitations and Data-Quality Notes

The dataset is strongly concentrated in Delhi-NCR, so smaller-city counts should not be interpreted as true global restaurant density. The currency field contains 13 unique currencies, so the overall cost value should be treated as a raw dataset measure unless currency normalization is performed. The grouping output also exposes anomalous non-standard values ('Orange' for online delivery and '3.2' for table booking) that should be validated in the source data.

The notebook additionally contains a correlation heatmap for Average Cost for two, Price range, Aggregate rating, and Votes. This is supplementary to the five required assignment charts.